

Watch 0 Fork 0 Star

Speech-to-text, text-to-speech, speaker diarization, speech enhancement, source separation, and VAD using next-gen Kaldi with onnxruntime without Internet connection. Support embedded systems, Android, iOS, HarmonyOS, Raspberry Pi, RISC-V, RK NPU, Axera NPU, Ascend NPU, x86_64 servers, websocket server/client, support 12 programming languages

Apache License 2.0
k2-fsa.github.io/sherpa-onnx/index.html
0 stars 0 forks 0 watching 1 branch 0 tags Activity
Public repository · Forked from k2-fsa/sherpa-onnx

m... 1 Branch 0 Tags Go to file Go to file Add file Code

This branch is up to date with k2-fsa/sherpa-onnx:master . Contribute Sync fork

nati2405	Add Paraformer Rust API example add rust paraformer example (k2-fsa#3707)	88bbc82 · 17 hours ago
.github	Add Paraformer Rust API example add rust ...	17 hours ago
android	Fix building Android KWS demo (k2-fsa#373...)	3 weeks ago
c-api-examples	Add API to get onnxruntime version. (k2-fsa...)	2 days ago
cmake	build(cmake): update spacemit-ort version (...)	2 days ago
cxx-api-examples	Support swift package manager (SPM) (k2-f...)	yesterday
dart-api-examples	Add API to get onnxruntime version. (k2-fsa...)	2 days ago
dotnet-examples	Add API to get onnxruntime version. (k2-fsa...)	2 days ago
ffmpeg-examples	Fix style issues (k2-fsa#1458)	2 years ago
flutter-examples	Release v1.13.4 (k2-fsa#3730)	3 weeks ago
flutter	Add API to get onnxruntime version. (k2-fsa...)	2 days ago
go-api-examples	Add API to get onnxruntime version. (k2-fsa...)	2 days ago
harmony-os	Add API to get onnxruntime version. (k2-fsa...)	2 days ago
ios-swift	Fix iOS examples using swift package mana...	17 hours ago
ios-swiftui	Fix iOS examples using swift package mana...	17 hours ago
java-api-examples	Add API to get onnxruntime version. (k2-fsa...)	2 days ago
kotlin-api-examples	Add API to get onnxruntime version. (k2-fsa...)	2 days ago
lazarus-examples	Fix typos in the project (k2-fsa#3293)	4 months ago
mfc-examples	Release v1.13.4 (k2-fsa#3730)	3 weeks ago
nodejs-addon-examples	Fix use-after-free in async TTS progress call...	last week
nodejs-examples	Release v1.13.4 (k2-fsa#3730)	3 weeks ago
pascal-api-examples	Add API to get onnxruntime version. (k2-fsa...)	2 days ago

 python-api-examples	Fix non_streaming_server.py crash when usi...	yesterday
 rust-api-examples	Add Paraformer Rust API example add rust ...	17 hours ago
 scripts	Publish an android-x64 .NET runtime packa...	yesterday
 sherpa-onnx	Fix streaming paraformer feature frontend t...	20 hours ago
 spm-examples	Support swift package manager (SPM) (k2-f...	yesterday
 swift-api-examples	Fix iOS examples using swift package mana...	17 hours ago
 tauri-examples	Release v1.13.4 (k2-fsa#3730)	3 weeks ago
 toolchains	Update openfst to v1.8.5 and fix compiler w...	3 months ago
 wasm	Export KittenTTS models (k2-fsa#3604)	2 months ago
 .clang-format	add java wrapper support (k2-fsa#117)	3 years ago
 .clang-tidy	Support clang-tidy (k2-fsa#1034)	2 years ago
 .flake8	add offline websocket server/client (k2-fsa#...	3 years ago
 .gitignore	Add Java API examples with gradle (k2-fsa#...	last week
 CHANGELOG.md	Release v1.13.4 (k2-fsa#3730)	3 weeks ago
 CMakeLists.txt	Release v1.13.4 (k2-fsa#3730)	3 weeks ago
 CPPLINT.cfg	Use static libraries for MFC examples (k2-fs...	3 years ago
 LICENSE	Use standard apache 2.0 license (k2-fsa#53)	3 years ago
 MANIFEST.in	Fix building wheels from source. (k2-fsa#632)	2 years ago
 Package.swift	Support swift package manager (SPM) (k2-f...	yesterday
 README.md	Add Domia to 'Projects using sherpa-onnx' (...)	last week
 build-aarch64-linux-gnu.sh	Fix building wheels (k2-fsa#2619)	10 months ago
 build-android-arm64-v8a.sh	Update onnxruntime to 1.27.0 (k2-fsa#3718)	27 days ago
 build-android-armv7-eabi.sh	Update onnxruntime to 1.27.0 (k2-fsa#3718)	27 days ago
 build-android-x86-64.sh	Update onnxruntime to 1.27.0 (k2-fsa#3718)	27 days ago
 build-android-x86.sh	Update onnxruntime to 1.27.0 (k2-fsa#3718)	27 days ago
 build-arm-linux-gnueabi.sh	Fix 32-bit arm CI (k2-fsa#2276)	last year
 build-axcl-linux-aarch64.sh	Refactor sense voice impl (k2-fsa#2873)	7 months ago
 build-axera-linux-aarch64.sh	Support AXERA ax630, ax650, and axcl back...	8 months ago
 build-ios-no-tts.sh	Support swift package manager (SPM) (k2-f...	yesterday
 build-ios-shared.sh	Release v1.13.4 (k2-fsa#3730)	3 weeks ago
 build-ios.sh	Support swift package manager (SPM) (k2-f...	yesterday
 build-ohos-arm64-v8a.sh	Publish sherpa_onnx.har for HarmonyOS (k...	2 years ago
 build-ohos-armeabi-v7a.sh	HarmonyOS support for VAD. (k2-fsa#1561)	2 years ago
 build-ohos-x86-64.sh	Publish sherpa_onnx.har for HarmonyOS (k...	2 years ago
 build-riscv64-linux-gnu-spacemit.sh	Update Spacemit toolchain download URL (...)	2 months ago

 build-riscv64-linux-gnu.sh	Build websocket related binaries for embed...	2 years ago
 build-rknn-linux-aarch64.sh	Fix building wheels (k2-fsa#2619)	10 months ago
 build-swift-macos.sh	Use SPM in iOS examples (k2-fsa#3822)	19 hours ago
 build-wasm-simd-asr.sh	WASM fixes (Bump emsdk 4.0.23) (k2-fsa#...)	2 months ago
 build-wasm-simd-kws.sh	WASM fixes (Bump emsdk 4.0.23) (k2-fsa#...)	2 months ago
 build-wasm-simd-nodejs.sh	WASM fixes (Bump emsdk 4.0.23) (k2-fsa#...)	2 months ago
 build-wasm-simd-speaker-diarization....	WASM fixes (Bump emsdk 4.0.23) (k2-fsa#...)	2 months ago
 build-wasm-simd-speech-enhanceme...	WASM fixes (Bump emsdk 4.0.23) (k2-fsa#...)	2 months ago
 build-wasm-simd-tts.sh	WASM fixes (Bump emsdk 4.0.23) (k2-fsa#...)	2 months ago
 build-wasm-simd-vad-asr.sh	WASM fixes (Bump emsdk 4.0.23) (k2-fsa#...)	2 months ago
 build-wasm-simd-vad.sh	WASM fixes (Bump emsdk 4.0.23) (k2-fsa#...)	2 months ago
 jitpack.yml	Test Java API on Windows arm64 (k2-fsa#3...)	last week
 new-release.sh	Refactor building and testing Java API (k2-f...)	last week
 pom.xml	Release v1.13.4 (k2-fsa#3730)	3 weeks ago
 release.sh	Publish pre-compiled libs for Android. (k2-fs...)	3 years ago
 setup.py	Release v1.13.4 (k2-fsa#3730)	3 weeks ago

README	License	
--------	---------	---

 Ask DeepWiki

Supported functions

Speech recognition	Speech synthesis	Source separation
✓	✓	✓

Speaker identification	Speaker diarization	Speaker verification
✓	✓	✓

Spoken Language identification	Audio tagging	Voice activity detection
✓	✓	✓

Keyword spotting	Add punctuation	Speech enhancement
✓	✓	✓

Supported platforms

Architecture	Android	iOS	Windows	macOS	linux	HarmonyOS
x64	✓		✓	✓	✓	✓
x86	✓		✓			
arm64	✓	✓	✓	✓	✓	✓
arm32	✓				✓	✓

Architecture	Android	iOS	Windows	macOS	linux	HarmonyOS
riscv64					✓	

Supported programming languages

1. C++	2. C	3. Python	4. JavaScript
✓	✓	✓	✓

5. Java	6. C#	7. Kotlin	8. Swift
✓	✓	✓	✓

9. Go	10. Dart	11. Rust	12. Pascal
✓	✓	✓	✓

It also supports WebAssembly.

Supported NPUs

1. Rockchip NPU (RKNN)	2. Qualcomm NPU (QNN)	3. Ascend NPU
✓	✓	✓

4. Axera NPU
✓

[Join our discord](#)

Introduction

This repository supports running the following functions **locally**

- Speech-to-text (i.e., ASR); both streaming and non-streaming are supported
- Text-to-speech (i.e., TTS)
- Speaker diarization
- Speaker identification
- Speaker verification
- Spoken language identification
- Audio tagging
- VAD (e.g., [silero-vad](#))
- Speech enhancement (e.g., [gtcrn](#), [DPDFNet](#))
- Keyword spotting
- Source separation (e.g., [spleeter](#), [UVR](#))

on the following platforms and operating systems:

- x86, x86_64, 32-bit ARM, 64-bit ARM (arm64, aarch64), RISC-V (riscv64), **RK NPU**, **Ascend NPU**
- Linux, macOS, Windows, openKylin
- Android, WearOS
- iOS
- HarmonyOS
- NodeJS
- WebAssembly
- [NVIDIA Jetson Orin NX](#) (Support running on both CPU and GPU)
- [NVIDIA Jetson Nano B01](#) (Support running on both CPU and GPU)

- [Raspberry Pi](#)
- [RV1126](#)
- [LicheePi4A](#)
- [VisionFive 2](#)
- [旭日X3派](#)
- [爱芯派](#)
- [RK3588](#)
- [SpacemiT-K1](#)
- [SpacemiT-K3](#)
- etc

with the following APIs

- C++, C, Python, Go, C#
- Java, Kotlin, JavaScript
- Swift, Rust
- Dart, Object Pascal

🔗 Links for Huggingface Spaces

- ▶ You can visit the following Huggingface spaces to try sherpa-onnx without installing anything. All you need is a browser.

🔗 Links for pre-built Android APKs

- ▶ You can find pre-built Android APKs for this repository in the following table

🔗 Links for pre-built Flutter APPs

- ▶ Details

🔗 Links for pre-built Lazarus APPs

- ▶ Details

🔗 Links for pre-trained models

- ▶ Details

🔗 Some pre-trained ASR models (Streaming)

- ▶ Details

🔗 Some pre-trained ASR models (Non-Streaming)

- ▶ Details

🔗 Useful links

- Documentation: <https://k2-fsa.github.io/sherpa/onnx/>
- Bilibili 演示视频: <https://search.bilibili.com/all?keyword=%E6%96%B0%E4%B8%80%E4%BB%A3Kaldi>

🔗 How to reach us

Please see <https://k2-fsa.github.io/sherpa/social-groups.html> for 新一代 Kaldi 微信交流群 and QQ 交流群.

🔗 Projects using sherpa-onnx

🔗 [Sherpa Voice / @siteed/sherpa-onnx.rn](#)

React Native wrapper and demo app for validating sherpa-onnx on iOS, Android, and Web, including ASR, TTS, VAD, KWS, speaker ID, diarization, language ID, punctuation, audio tagging, and speech enhancement.

- [NPM package](#)
- [Live demo](#)

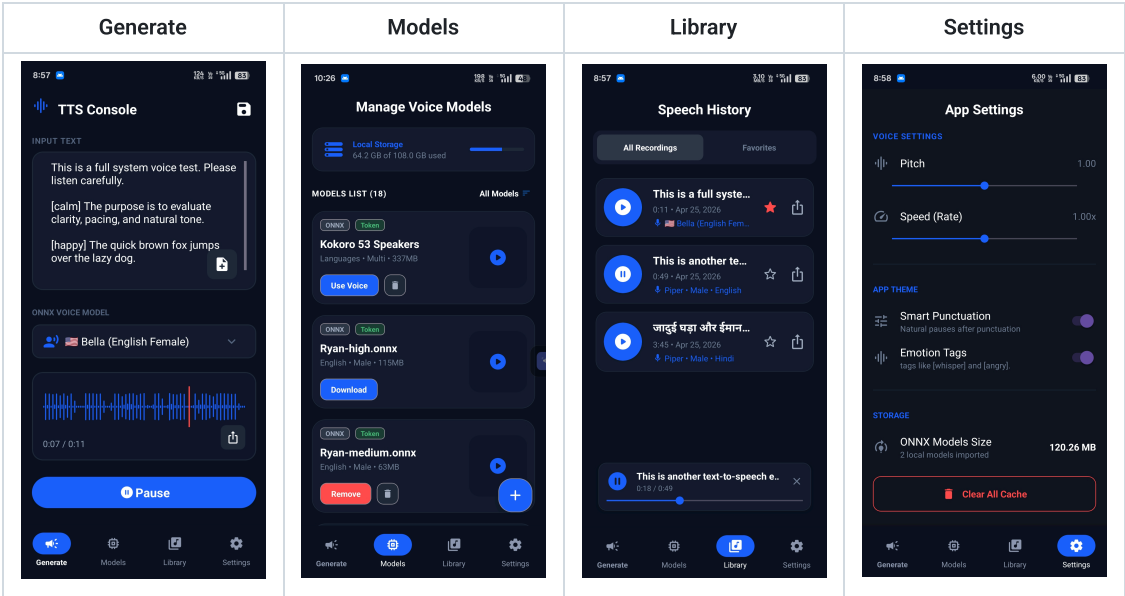
🔗 **Speed of Sound**

A voice-typing application for the Linux desktop (GTK4/Adwaita). It captures microphone audio, transcribes it offline using Sherpa ONNX ASR models, optionally polishes the text with an LLM, and types the result into the active window via XDG Remote Desktop Portal keyboard simulation.

🔗 **VoxSherpa TTS**

VoxSherpa TTS is a 100% offline Android Text-to-Speech app powered by Sherpa-ONNX. It supports Kokoro-82M, Piper, and VITS engines with multilingual support including Hindi, English, British English, Japanese, Chinese and 50+ more languages.

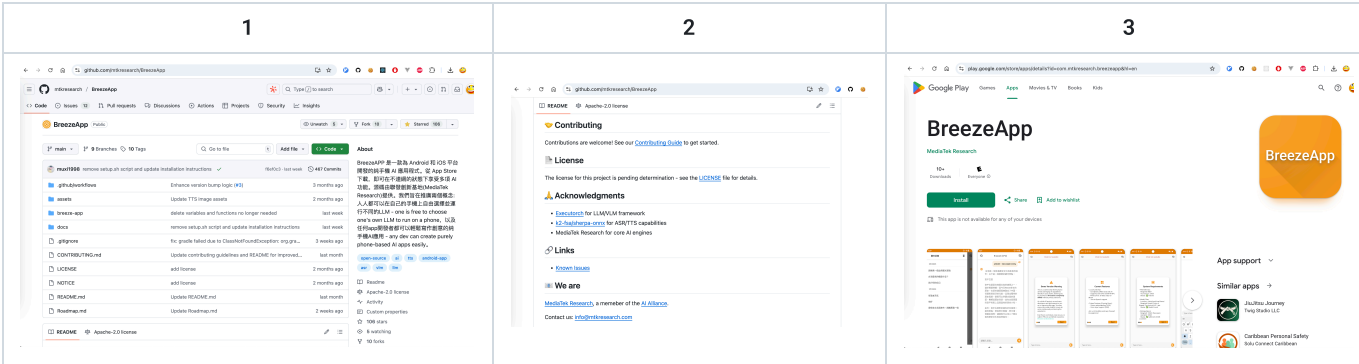
- [Download APK \(All Versions\)](#)
- Android 11+ · 100% offline · No telemetry



🔗 **BreezeApp from MediaTek Research**

BreezeAPP is a mobile AI application developed for both Android and iOS platforms. Users can download it directly from the App Store and enjoy a variety of features offline, including speech-to-text, text-to-speech, text-based chatbot interactions, and image question-answering

- [Download APK for BreezeAPP](#)
- [APK 中国镜像](#)



🔗 **Open-LLM-VTuber**

Talk to any LLM with hands-free voice interaction, voice interruption, and Live2D taking face running locally across platforms

See also <https://github.com/t41372/Open-LLM-VTuber/pull/50>

🔗 [voiceapi](#)

► Streaming ASR and TTS based on FastAPI

🔗 [腾讯会议摸鱼工具 TMSpeech](#)

Uses streaming ASR in C# with graphical user interface.

Video demo in Chinese: [【开源】Windows实时字幕软件（网课/开会必备）](#)

🔗 [lol互动助手](#)

It uses the JavaScript API of sherpa-onnx along with [Electron](#)

Video demo in Chinese: [爆了！炫神教你开打字挂！真正影响胜率的英雄联盟工具！英雄联盟的最后一块拼图！和游戏中的每个人无障碍沟通！](#)

🔗 [Sherpa-ONNX 语音识别服务器](#)

A server based on nodejs providing Restful API for speech recognition.

🔗 [QSmartAssistant](#)

一个模块化，全过程可离线，低占用率的对话机器人/智能音箱

It uses QT. Both [ASR](#) and [TTS](#) are used.

🔗 [Flutter-EasySpeechRecognition](#)

It extends [./flutter-examples/streaming_asr](#) by downloading models inside the app to reduce the size of the app.

Note: [\[Team B\] Sherpa AI backend](#) also uses sherpa-onnx in a Flutter APP.

🔗 [sherpa-onnx-unity](#)

sherpa-onnx in Unity. See also [#1695](#), [#1892](#), and [#1859](#)

🔗 [xiaozi-esp32-server](#)

本项目为xiaozi-esp32提供后端服务，帮助您快速搭建ESP32设备控制服务器 Backend service for xiaozi-esp32, helps you quickly build an ESP32 device control server.

See also

- [ASR新增轻量级sherpa-onnx-asr](#)
- [feat: ASR增加sherpa-onnx模型](#)

🔗 [KaithemAutomation](#)

Pure Python, GUI-focused home automation/consumer grade SCADA.

It uses TTS from sherpa-onnx. See also [🌟 Speak command that uses the new globally configured TTS model.](#)

🔗 [Open-XiaoAI KWS](#)

Enable custom wake word for XiaoAi Speakers. 让小爱音箱支持自定义唤醒词。

Video demo in Chinese: [小爱同学启动～„！ㄊ！„！](#)

🔗 [C++ WebSocket ASR Server](#)

It provides a WebSocket server based on C++ for ASR using sherpa-onnx.

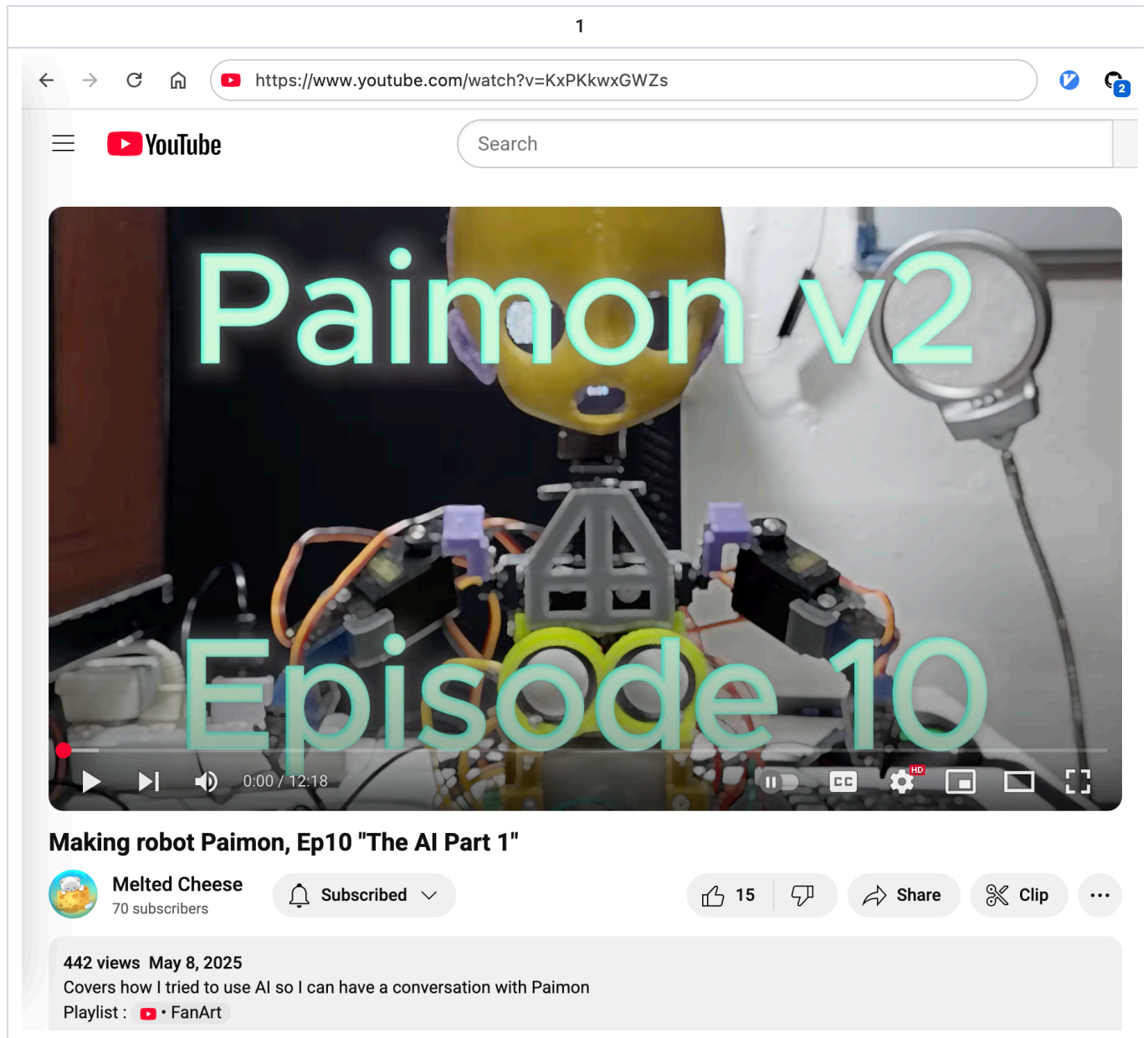
🔗 [Go WebSocket Server](#)

It provides a WebSocket server based on the Go programming language for sherpa-onnx.

🔗 [Making robot Paimon, Ep10 "The AI Part 1"](#)

It is a [YouTube video](#), showing how the author tried to use AI so he can have a conversation with Paimon.

It uses sherpa-onnx for speech-to-text and text-to-speech.



🔗 [TtsReader - Desktop application](#)

A desktop text-to-speech application built using Kotlin Multiplatform.

🔗 [MentraOS](#)

Smart glasses OS, with dozens of built-in apps. Users get AI assistant, notifications, translation, screen mirror, captions, and more. Devs get to write 1 app that runs on any pair of smart glasses.

It uses sherpa-onnx for real-time speech recognition on iOS and Android devices. See also [Mentra-Community/MentraOS#861](#)

It uses Swift for iOS and Java for Android.

🔗 [flet_sherpa_onnx](#)

Flet ASR/STT component based on sherpa-onnx. Example [a chat box agent](#)

🔗 [achatbot-go](#)

a multimodal chatbot based on go with sherpa-onnx's speech lib api.

🔗 [fcitx5-vinput](#)

Local offline voice input plugin for [Fcitx5](#) (Linux input method framework). It uses C++ with offline ASR for speech recognition, supporting push-to-talk, command mode, and optional LLM post-processing.

Video demo in Chinese: [fcitx5-vinput](#)

🔗 [Wake Word](#)

A VS Code extension for hands-free voice-activated coding. It uses sherpa-onnx for real-time keyword spotting (KWS) to detect custom wake phrases and trigger VS Code commands by voice. Audio capture is handled by [decibri](#), a cross-platform Node.js microphone streaming library with prebuilt native binaries.

- [VS Code Marketplace](#)
- [Open VSX](#)
- [decibri integration guides for sherpa-onnx](#)

🔗 [SmartSub](#)

SmartSub is a local-first cross-platform desktop application for the complete subtitle production pipeline: audio/video transcription, subtitle translation, proofreading, and subtitle burning/muxing.

It natively integrates sherpa-onnx to power three offline ASR engines — FunASR, Qwen3-ASR, and FireRedASR — delivering high-accuracy Chinese and multilingual speech recognition entirely on-device, with no file uploads required.

🔗 [Fono](#)

Talk to your computer. Fono is an open-source (GPL-3.0), complete voice-AI stack (speech-to-text, text-to-speech, a local LLM, wake word, speaker ID) in one small binary for the Linux, macOS, and Windows desktop. Press a key and speak: it types into any app, answers as a voice assistant, or drives a coding agent.

It uses sherpa-onnx's Supertonic TTS models for text-to-speech, with the Supertonic text frontend ported to Rust. See also [#3750](#), which fixes dropped diacritics in the Supertonic frontend for Romanian, Czech, Polish, Hungarian, Turkish, Vietnamese, and other non-English Latin-script languages.

🔗 [Domia](#)

Domia is an open-source local AI companion you talk to, living entirely on hardware you own. Each Domia is its own character, with a distinct personality, voice, emotional state, and memory of you and its place — and it acts, not just talks: skills connect it to the real world over MCP, Home Assistant included. Domia nodes form a local peer-to-peer mesh that shares speech and reasoning compute, while room satellites bring each companion into the spaces where you live without giving up its identity. No cloud, no accounts, and no data leaving your network.

Domia runs sherpa-onnx in-process for keyword spotting, voice activity detection, streaming and offline speech recognition, and text-to-speech. Speech engines and models are selected through per-companion configuration, making it easy to match supported sherpa-onnx

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors

No contributors

Languages



Suggested workflows

Based on your tech stack



CMake based, multi-platform projects
Build and test a CMake based project on multiple platforms.
By GitHub Actions

Configure



CMake based, single-platform projects
Build and test a CMake based project on a single-platform.
By GitHub Actions

Configure



Publish Python Package
Publish a Python Package to PyPI on release.
By GitHub Actions

Configure

[More workflows](#)

